

Learning the Routing Policy for VIA-SD: RL-Gated Hierarchical Speculative Decoding

viasd-rl experiments

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Abstract

We reimplement VIA-SD (arXiv:2606.12243) — speculative decoding with a slim, layer-skipped intermediate verifier q' — and replace its hand-tuned (α_1, α_2) confidence thresholds with a **learned per-token gating policy**. We study three RL formulations (REINFORCE, GRPO, and a per-token-reward variant) and their reward/credit-assignment design. On a GSM8K slice with a Qwen2.5 0.5B→14B pair, the learned gate **dramatically outperforms the paper’s fixed thresholds** (accuracy 0.21–0.40 → 0.88–0.91) at fewer expensive verifier calls. We also report honest negatives: plain lossless speculative decoding stays Pareto-competitive at this gap, and the slim-verifier tier only pays off when the gate is learned.

1 Background: VIA-SD and DIMR

Speculative decoding (SD). A cheap *drafter* p autoregressively proposes a block of γ candidate tokens; the expensive *target* verifier q scores all γ in a single parallel forward and accepts the longest prefix consistent with its own distribution, recomputing the first rejected token. This is *lossless* (output equals sampling/greedy from q) and trades extra compute for fewer *sequential* large-model steps. Its single lever is the *acceptance rate*: each rejection truncates the block and forces a full-model step.

VIA-SD’s contribution. The paper observes that many rejected tokens lie in a “middle zone” — the drafter is close but not fully reliable — and do not actually need the full model. It inserts a *slim intermediate verifier* q' between p and q , turning the binary accept/reject into a three-way *hierarchical* decision per token: **accept** p ’s token when q' is highly confident; **regenerate** with q' for medium-confidence (middle-zone) tokens; **escalate** to the full q only for low-confidence tokens. By resolving middle-zone tokens at q' , it reduces how often the expensive q is invoked relative to plain SD. The routing is governed by two fixed confidence thresholds (α_1, α_2) on q' ’s confidence ratio (reported best at $\alpha_1=0.5, \alpha_2=0.3$); the scheme is mildly *lossy*, with a tolerance α controlling how far accepted tokens may deviate. The paper motivates the intermediate stage information-theoretically: in KL geometry an intermediate distribution can lower the *cumulative* divergence on the path $p \rightarrow q' \rightarrow q$ below the direct $p \rightarrow q$, whereas total-variation (which exactly equals the lossless rejection rate) forbids this.

DIMR (Dynamic Intra-Model Routing): how q' is built. Rather than a separate model, q' is a *routed sub-model of q itself*: a binary mask $z \in \{0, 1\}^L$ selects which of q ’s L decoder layers to keep, $q' = \Pi_z(q)$, skipping the rest (each decoder block is residual, so dropping it is an identity on the residual stream — dimensionally safe). q' *shares* q ’s embeddings and output head, so it adds no parameters and only changes inference dynamics. DIMR is the *offline search* for the mask z^* : it minimizes an accumulated KL-style cost between q and q'_z over a calibration

window (the paper’s Eq. 12), combining random search with Bayesian/greedy refinement, keeping first/last layers and skipping $\sim 45\%$ of the middle. The cost is *not* raw $\text{KL}(q\|q')$ but the Eq. 11 *margin-violation* cost — a two-term ReLU penalty (acceptance-threshold + residual-replacement, at (α, β)) that is the quantity actually governing the $q' \rightarrow q$ gate’s rejection rate. The paper shows DIMR-selected masks beat random/ evenly-spaced ones; our faithful search confirms this ($\sim 50\%$ lower cost than evenly-spaced).

2 Methodology (our extension)

Pipeline. Drafter p (0.5B) proposes a $\gamma=5$ -token block; for each token a gate chooses one of three actions: *accept* p ’s token (cheap), *regenerate* with the slim verifier q' (a layer-skipped copy of the full verifier $q=14\text{B}$, medium), or *escalate* to q (expensive). The block commits up to the first changed token, then redrafts.

Learned gate. The fixed (α_1, α_2) thresholds are replaced by a tiny MLP over q -free features (drafter/ q' entropy, top-1 prob, confidence ratio $q'(v)/\max q'$, p/q' agreement, $\text{KL}(p\|q')$, block position). Crucially, q is *never* a policy input — it is used only at training time to compute rewards/labels — so the gate is deployable without running the full model to decide.

Slim verifier q' / DIMR. q' skips $\sim 45\%$ of q ’s decoder layers (residual blocks $\Rightarrow$ identity, dimensionally safe). The skip mask is found by an offline DIMR search minimizing the paper’s Eq. 11 *margin-violation* cost between q and q' (the two-term ReLU acceptance/residual cost, not raw KL); the searched mask is $\sim 50\%$ lower-cost than an evenly-spaced one.

Metrics. q -calls/token (full-verifier invocations per token; exact, hardware-independent), GSM8K accuracy, and speedup. SD’s win is latency, but measured wall-clock is dominated by a ~ 10 ms eager per-call overhead floor; we report an **overhead-free bandwidth-model speedup** (per-forward cost = active-params $\times 2$ bytes/HBM-bw), cross-validated by a measure-and-subtract estimate (they agree for single-tier methods).

3 Reinforcement Learning for the Gating Policy

3.1 Problem framing

Each drafted-and-gated token is a decision step: state s_t = the q -free feature vector, action $a_t \in \{\text{ACCEPT}, \text{REGEN}, \text{ESCALATE}\}$, policy π_θ a 2-layer MLP. Routing effects are largely *local* (the action at t sets token t and perturbs nearby context), which motivates a contextual-bandit view. All policies are **warm-started from an imitation policy** (below); RL then refines on the policy’s *own* rollout distribution.

3.2 Imitation warm-start and exposure bias

An oracle that may consult q labels the cost-minimal route reproducing greedy- q (ACCEPT if p ’s token = $\arg \max q$; REGEN if $\arg \max q' = \arg \max q$; else ESCALATE); the MLP is fit with class-weighted cross-entropy. Imitation alone reaches only ~ 0.33 GSM8K accuracy: trained on oracle-visited (near-greedy) states, it over-accepts at its *own* states where errors compound — classic **exposure bias**. RL’s primary value here is fixing this by training on-policy, lifting accuracy to 0.88+.

3.3 Reward design

The decoding goal is to reproduce the target model’s output cheaply, giving two ingredients: *quality* and *cost*.

- **Quality (dense proxy).** $\text{match}_t = \mathbf{1}[\text{emitted}_t = \arg \max q(\cdot \mid x_{<t})]$, computed by running q at every position *at training time only* (not charged to cost).
- **Cost.** Per-action latency in *overhead-corrected* units: $\text{cost}_t = (t_{p1} + t_{q'} / \gamma + \mathbf{1}[\text{ESCALATE}] t_q) / t_{q1}$ — accept ≈ 0.1 , escalate ≈ 1.5 .
- **Correctness (sparse, true objective).** $\text{correct} \in \{0, 1\}$, GSM8K final-answer match.

We compare three reward forms:

$$\text{REINFORCE/GRPO (trajectory): } R = \text{match_rate} + r_c \text{correct} - \lambda \overline{\text{cost}} \quad (1)$$

$$\text{v2 (per-token): } r_t = \text{match}_t - \lambda \text{cost}_t + \frac{r_c \text{correct}}{T} \quad (2)$$

$$\text{correctness-focused GRPO: } R = \text{correct} - \lambda \overline{\text{cost}} \quad (3)$$

The proxy gap. match is a *dense surrogate* for the *sparse* true objective (correctness). We measured the disconnect: a policy matching greedy- q on 98.5% of tokens still scores only ~ 0.72 GSM8K accuracy, because a few deviations per chain derail the final answer. This motivates the correctness-focused reward (Eq. 3), which optimizes the true objective directly; its ceiling is bounded by the q -free features (a confidently-*wrong* drafter token looks safe without q). A cost penalty is retained or it collapses to “escalate everything” = greedy.

3.4 Credit assignment: per-trajectory vs. per-token

Trajectory-level (REINFORCE, GRPO). One scalar R per rollout is applied to *all* ~ 150 routing decisions: $\nabla_{\theta} J = E[A \sum_t \nabla \log \pi(a_t \mid s_t)]$. Crude — a good and a bad decision in the same rollout receive identical credit — and high-variance. **Per-token (v2).** Each decision is treated as a local bandit step with its *own* r_t and advantage, exploiting the locality of routing. This is far lower-variance and more precise; empirically it produces the smoothest training curves (Fig. 3) and the lowest cost/token.

3.5 Advantage estimation: REINFORCE vs. GRPO

REINFORCE uses $A = R - b$ with b a *global EMA* of reward across *all* prompts. This conflates per-*problem* difficulty (easy prompts give high R regardless of routing) with policy quality, injecting variance. **GRPO** samples a *group* of G rollouts of the *same* prompt and sets $A_i = (R_i - \text{mean}_{\text{group}}) / (\text{std}_{\text{group}} + \epsilon)$, cancelling problem difficulty with no value network; it adds a PPO-clipped surrogate (clip 0.2, $K=3$ inner epochs), a KL penalty to the frozen warm-start reference, and an entropy bonus. GRPO is the right tool for the *sparse correctness* reward (Eq. 3): the group baseline yields a gradient precisely on the mixed-difficulty prompts where some of the G rollouts succeed and others fail. **v2** uses batch-normalized per-token advantages (contextual-bandit REINFORCE) with an entropy bonus and gradient clipping.

3.6 What we observed

Fig. 3: **v2 per-token** is the lowest-variance and drives rejection lowest; **GRPO** is noisiest (group-level binary-ish reward); **REINFORCE** is intermediate; all converge to $\text{match} \approx 0.95$. On the

efficiency metric, GRPO pushed q-calls/token lowest among the match-rewarded variants (0.162), while the v2 reward drove escalation to 0.12 (fastest). Because the match-based rewards are match-dominated, the resulting RL is close to *on-policy imitation* (it fixes exposure bias rather than discovering exotic routing); the correctness-focused GRPO (Eq. 3) is the formulation that targets the true objective directly.

4 Results

Method	GSM8K acc	q-calls/tok	speedup (bw)
greedy- q (reference)	0.887	1.000	1.00×
plain SD (lossless)	0.907	0.259	3.30×
via_fixed (paper, tuned)	0.40 [*]	0.31	1.76×
via_imit (imitation only)	0.34	0.29	1.69×
via_rl REINFORCE	0.880	0.243	—
via_rl GRPO	0.853	0.162	—
via_rl REINFORCE + DIMR	0.907	0.250	2.22×
via_rl v2 per-token ($\lambda=0.3$)	0.713	0.104	3.53×

Table 1: Headline comparison (0.5B→14B, GSM8K). ^{*}via_fixed peaks at ≈ 0.50 even under a full 9-config threshold sweep. q-calls/token is exact; bandwidth speedup reported where measured. Accuracy uses $n=80\text{--}150$ (see caveats).

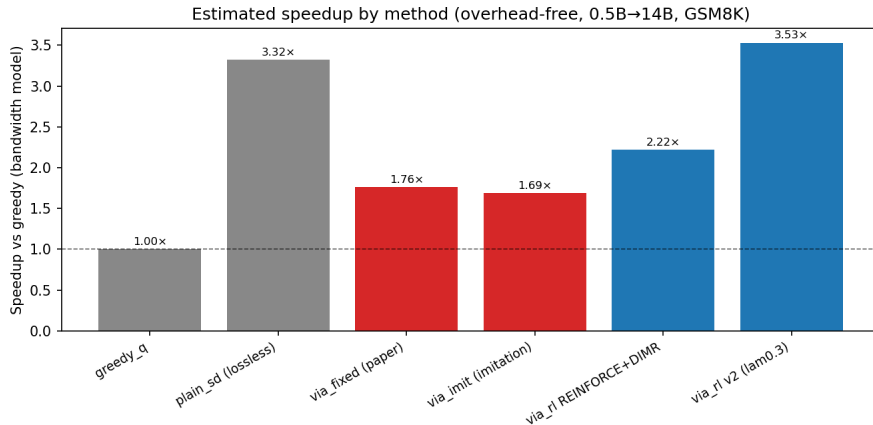


Figure 1: Overhead-free speedup by method. The learned v2 gate and plain SD reach $\sim 3.3\text{--}3.5\times$; the fixed-threshold gate is slow because it over-escalates to the full model.

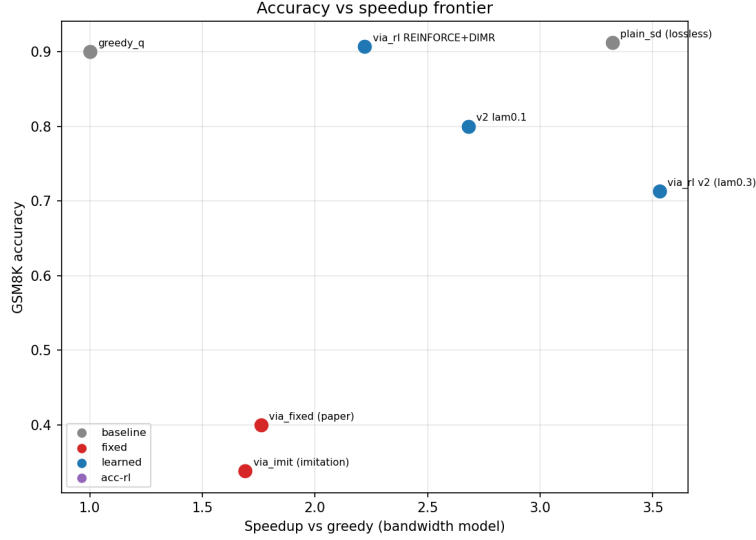


Figure 2: Accuracy vs. speedup. **Learned** gates dominate **fixed** ones; REINFORCE+DIMR matches plain SD’s accuracy (0.907) at lower speed (the q' overhead); v2 trades accuracy for speed.

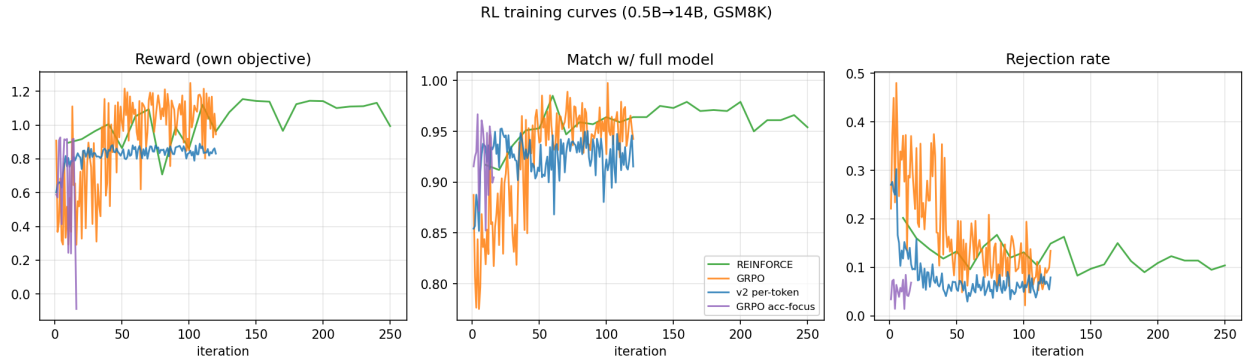


Figure 3: RL training curves. v2 per-token (blue) is lowest-variance and drives rejection lowest; GRPO (orange) is noisiest; REINFORCE (green) intermediate; all reach match $\approx$ 0.95.

5 Findings

1. **Learned gate** $\gg$ **fixed thresholds** (0.88–0.91 vs. 0.21–0.40, and ≤ 0.50 fully tuned), at lower q -calls/token — robust across masks and capacity gaps.
2. **RL fixes imitation’s exposure bias** (0.33 $\rightarrow$ 0.88+) by training on-policy.
3. **DIMR helps only the learned gate** (via.rl 0.88 $\rightarrow$ 0.91); the fixed rule stays brittle.
4. **Credit assignment matters**: per-token (v2) is far lower-variance than trajectory-level REINFORCE/GRPO and reaches the lowest escalation; GRPO’s group baseline beats REINFORCE’s global one.
5. **v2 unlocks VIA-SD > plain-SD on speed** ($3.53\times > 3.30\times$, escalation below break-even) but overshoots accuracy (0.71) at $\lambda=0.3$.

6. **Honest negatives:** plain lossless SD stays Pareto-competitive (the q' tier overhead doesn't pay off at this gap); a bigger *verifier* (32B) hurts an accept-heavy policy (acc 0.33) — a bigger *drafter* is the indicated lever.

6 Caveats

Accuracy at small n (24–50) carries a ± 0.06 –0.10 CI, so small accuracy gaps are within noise (large ones are not). Speedups are an overhead-free cost *model*, not a stopwatch on an optimized engine (CUDA-graphs/static-cache were incompatible with the dynamic KV-cache loop). Single task (GSM8K), single model family (Qwen2.5).